

ATM Management System — Java Code

Main.java

```
import java.util.ArrayList;
import java.util.Scanner;

class Account {
    private int accountNumber;
    private String name;
    private int pin;
    private double balance;
    private ArrayList<String> transactions;

    public Account(int accountNumber, String name, int pin, double balance) {
        this.accountNumber = accountNumber;
        this.name = name;
        this.pin = pin;
        this.balance = balance;
        this.transactions = new ArrayList<>();
    }

    public int getAccountNumber() {
        return accountNumber;
    }

    public String getName() {
        return name;
    }

    public boolean checkPin(int enteredPin) {
        return pin == enteredPin;
    }

    public double getBalance() {
        return balance;
    }

    public void deposit(double amount) {
        balance += amount;
        transactions.add("Deposited: Rs. " + amount);
    }

    public boolean withdraw(double amount) {
        if (amount > balance) {
            return false;
        }

        balance -= amount;
        transactions.add("Withdrawn: Rs. " + amount);
        return true;
    }

    public void changePin(int newPin) {
        pin = newPin;
    }

    public void showTransactions() {
        System.out.println("\n===== MINI STATEMENT =====");

        if (transactions.isEmpty()) {
            System.out.println("No transactions yet.");
        } else {
            for (String transaction : transactions) {
                System.out.println(transaction);
            }
        }

        System.out.println("=====");
    }
}

public class Main {

    static Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {

        Account account = new Account(
```

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        1001,
        "Student",
        1234,
        10000.00
    );

    System.out.println("=====");
    System.out.println("        ATM MANAGEMENT SYSTEM");
    System.out.println("=====");

    // Login
    System.out.print("Enter Account Number: ");
    int accountNumber = scanner.nextInt();

    if (accountNumber != account.getAccountNumber()) {
        System.out.println("Invalid account number.");
        return;
    }

    boolean loggedIn = false;

    for (int attempt = 1; attempt <= 3; attempt++) {

        System.out.print("Enter PIN: ");
        int pin = scanner.nextInt();

        if (account.checkPin(pin)) {
            loggedIn = true;
            break;
        } else {
            System.out.println("Incorrect PIN.");

            if (attempt < 3) {
                System.out.println(
                    "Attempts remaining: " + (3 - attempt)
                );
            }
        }
    }

    if (!loggedIn) {
        System.out.println("Too many failed attempts.");
        System.out.println("Account access denied.");
        return;
    }

    System.out.println("\nLogin successful!");
    System.out.println("Welcome, " + account.getName());

    // ATM Menu
    while (true) {

        System.out.println("\n===== ATM MENU =====");
        System.out.println("1. Check Balance");
        System.out.println("2. Deposit Money");
        System.out.println("3. Withdraw Money");
        System.out.println("4. Mini Statement");
        System.out.println("5. Change PIN");
        System.out.println("6. Exit");
        System.out.println("=====");

        System.out.print("Enter your choice: ");
        int choice = scanner.nextInt();

        switch (choice) {

            case 1:
                // Check balance
                System.out.println("\n===== BALANCE =====");
                System.out.println(
                    "Account Number: "
                        + account.getAccountNumber()
                );
                System.out.println(
                    "Available Balance: Rs. "
                        + account.getBalance()
                );
                System.out.println("=====");
                break;

            case 2:
                // Deposit
                System.out.print(
                    "\nEnter amount to deposit: Rs. "

```

```

    );
    double depositAmount = scanner.nextDouble();

    if (depositAmount <= 0) {
        System.out.println(
            "Amount must be greater than zero."
        );
    } else {
        account.deposit(depositAmount);

        System.out.println(
            "Deposit successful!"
        );
        System.out.println(
            "New Balance: Rs. "
                + account.getBalance()
        );
    }
    break;

case 3:
    // Withdraw
    System.out.print(
        "\nEnter amount to withdraw: Rs. "
    );
    double withdrawAmount = scanner.nextDouble();

    if (withdrawAmount <= 0) {
        System.out.println(
            "Amount must be greater than zero."
        );
    } else if (account.withdraw(withdrawAmount)) {
        System.out.println(
            "Withdrawal successful!"
        );
        System.out.println(
            "Remaining Balance: Rs. "
                + account.getBalance()
        );
    } else {
        System.out.println(
            "Insufficient balance!"
        );
        System.out.println(
            "Available Balance: Rs. "
                + account.getBalance()
        );
    }
    break;

case 4:
    // Mini statement
    account.showTransactions();
    break;

case 5:
    // Change PIN
    System.out.print(
        "\nEnter current PIN: "
    );
    int currentPin = scanner.nextInt();

    if (!account.checkPin(currentPin)) {
        System.out.println("Incorrect current PIN.");
        break;
    }

    System.out.print("Enter new 4-digit PIN: ");
    int newPin = scanner.nextInt();

    if (newPin < 1000 || newPin > 9999) {
        System.out.println(
            "PIN must contain exactly 4 digits."
        );
    } else {
        account.changePin(newPin);
        System.out.println(
            "PIN changed successfully!"
        );
    }
    break;

case 6:
    // Exit

```

```
        System.out.println(
            "\nThank you for using ATM Management System."
        );
        System.out.println("Have a nice day!");
        scanner.close();
        return;

    default:
        System.out.println(
            "Invalid choice. Please try again."
        );
    }
}
}
```